



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

**CANDIDATE
NAME**

CLASS

CHEMISTRY
JC2 Preliminary Examination
Paper 3 Free Response

9729/03
20 Sept 2017
2 hours

Candidates answer on separate paper.

Additional materials : Answer Paper
Data Booklet
2 Cover Pages

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough work.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part question.

This document consists of **11** printed pages and **1** blank page.

Section A

Answer **all** the questions in this section.

- 1 (a) Oxides of nitrogen, NO_x are a mixture of gases that are composed of nitrogen and oxygen. They are formed during combustion processes in power plants.

Thermal NO_x refers to NO_x formed through high temperature oxidation of the diatomic nitrogen found in combustion air.

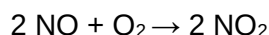
The formation rate is a function of temperature and the residence time of nitrogen at that temperature. At high flame temperatures, molecular nitrogen and oxygen dissociate into their atomic states and participate in a series of reactions.

The two principal reactions producing thermal nitric oxide, NO are as follows.

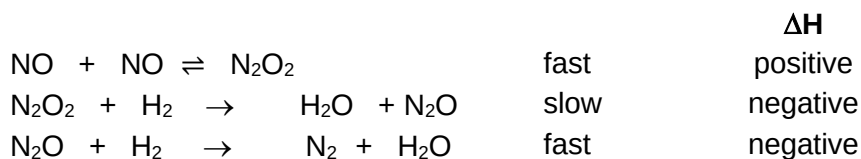
Step 1 O atoms react reversibly with N_2 to form NO.

Step 2 N atoms liberated react with O_2 to produce more NO.

Nitrogen dioxide typically arises via the oxidation of nitric oxide by oxygen in the atmosphere.

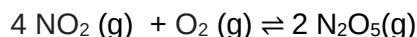


- (i) Write the two principal reactions involved in the formation of thermal nitric oxide in combustion. [1]
- (ii) Hence, determine the rate determining step. [1]
- (iii) Suggest a method to reduce the formation of thermal NO_x . [1]
- (b) Catalytic converters are used to reduce the output of NO_x into the environment. One example is the removal of nitric oxide by reaction with hydrogen gas using platinum catalyst.



- (i) Write the rate equation for the above reaction, showing clearly how you derive it. [2]
- (ii) Draw the probable energy profile diagram for this overall exothermic reaction. [2]
- (iii) Using an appropriate sketch of the Maxwell–Boltzmann distribution curve, explain how the addition of platinum affects the rate of chemical reaction. [3]

- (c) The NO_2 reacts with O_2 in the atmosphere to form colourless nitrogen pentoxide, N_2O_5 in equilibrium.



Dinitrogen pentoxide is a rare example of a compound that adopts two structures depending on the conditions. As a gas in the atmosphere, dinitrogen pentoxide exists as a molecule. However, it exists as an ionic salt in the solid state.

- (i) Gaseous N_2O_5 is a symmetrical molecule with the two nitrogen atoms bonded to one central oxygen atom.

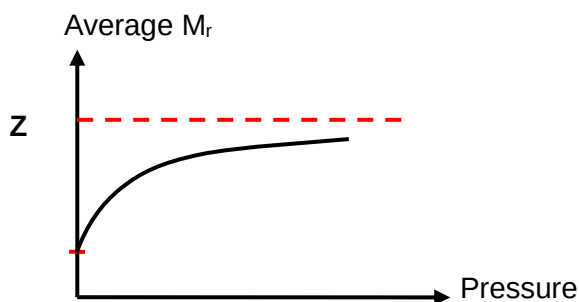
Draw a dot-and-cross diagram to show the bonding in N_2O_5 molecule. Use your diagram to:

- State the shape and the bond angle with respect to nitrogen
- Suggest with reasoning, the polarity of the molecule. [3]

- (ii) A ratio of 4 to 1 mole of gaseous NO_2 and O_2 with a total initial pressure of 2.25 atm was placed in a 1000 cm^3 vessel. The total pressure at equilibrium was 1.5 atm. Calculate K_p for this equilibrium. [3]

- (iii) Calculate the average M_r of the gases at equilibrium. [1]

- (iv) The variation of M_r is as shown in the sketch.



Predict a value for **Z** and account for the shape of the graph. [2]

- (d) Solid N_2O_5 dissolved in chloroform may be used to replace concentrated nitric acid and sulfuric acid in the nitration of benzene.

- (i) Suggest why solid N_2O_5 can be used in for nitration of benzene. [1]

- (ii) 0.60 mol of solid N_2O_5 was dissolved in chloroform. Some N_2O_5 decompose back into oxygen and nitrogen dioxide with a rate constant of $1.0 \times 10^{-5} \text{ s}^{-1}$ at 45°C .

The gases produced are collected in a 10 dm^3 container over a period of 40 hours. Calculate the resultant pressure.

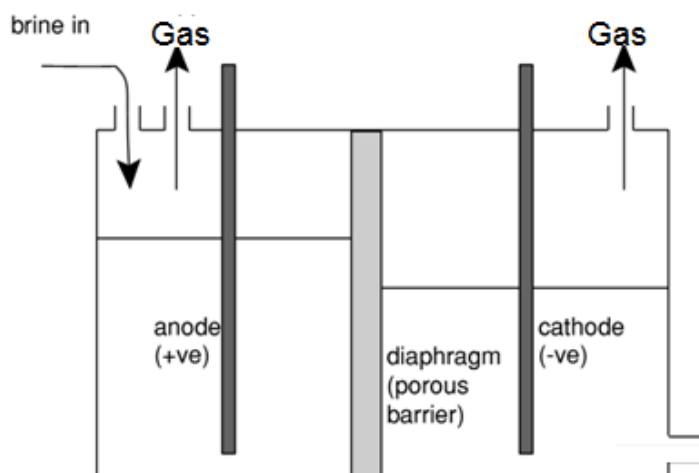
[Assume that the products do not dissolve in chloroform.] [3]

[Total : 23]

- 2 Chlorine is a yellow-green gas that was first synthesised around 1630. Chlorine forms a variety of oxides such as chlorate(VII), ClO_4^- and chlorate(V), ClO_3^- .

- (a) Chlorine gas can be produced from the electrolysis of brine (concentrated sodium chloride solution) using a diaphragm cell with inert electrodes.

A diaphragm is used to keep the products of electrolysis from mixing and hence, prevent the formation of undesirable by-products.

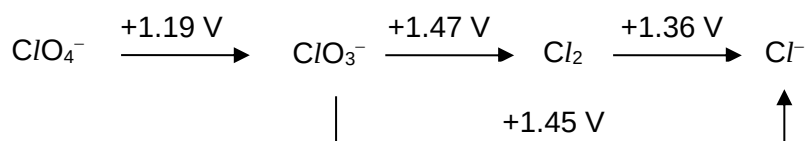


- (i) Write the ion-electron half-equations, with state symbols, for the reactions taking place at the electrodes. [2]
- (ii) Calculate the mass of chlorine gas produced if a current of 5 A was passed for 7 hours through the electrolytic cell. [2]
- (b) In some industrial electrolytic cells, the products of electrolysis of brine are allowed to react further when the diaphragm is removed. Two chlorine-containing products are formed, depending on the operating conditions used. At low temperatures, one of the products is a chloro-oxoanion with an oxidation state of +1.
- (i) State the type of reaction taking place. [1]
- (ii) Give an overall balanced equation for the reaction described above. [1]
- (c) With reference from the Data Booklet, explain why dilute aqueous sodium chloride cannot be used as the electrolyte for the manufacture of chlorine in this electrolysis. [1]

- (d) The standard electrode potentials, $E^\ominus$, and standard Gibbs free energy changes, $\Delta G^\ominus$, of different chlorine-containing species are tabulated below.

	Half-equation	$E^\ominus / \text{V}$	$\Delta G^\ominus / \text{kJ mol}^{-1}$
1	$\text{ClO}_4^- + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{ClO}_3^- + \text{H}_2\text{O}$	+1.19	-230
2	$2\text{ClO}_3^- + 12\text{H}^+ + 10\text{e}^- \rightleftharpoons \text{Cl}_2 + 6\text{H}_2\text{O}$	+1.47	?
3	$\text{Cl}_2 + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-$	+1.36	-262

These electrode potentials can be summarised using the Latimer diagram shown below. In a Latimer diagram, the most highly oxidised form of the element is on the left, with successively lower oxidation states to the right. The electrode potentials are shown on the arrows.



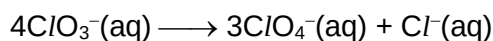
- (i) Calculate $\Delta G^\ominus$ for half-equation 2. [1]
- (ii) The standard electrode potential of converting ClO_3^- to Cl^- is found to be +1.45 V, instead of +2.83 V. The number of electrons transferred in each step must be taken into account.

I: Write a half-equation for the conversion of ClO_3^- to Cl^-

II: Using your knowledge of Hess' Law for $\Delta G^\ominus$ and your answer to (d)(i), show with the aid of an energy cycle that the $E^\ominus$ for the conversion of ClO_3^- to Cl^- is +1.45 V.

[4]

- (iii) With the help of the Latimer diagram provided, calculate $E^\ominus_{\text{cell}}$ of the reaction below. Hence, determine its $\Delta G^\ominus_{\text{cell}}$.



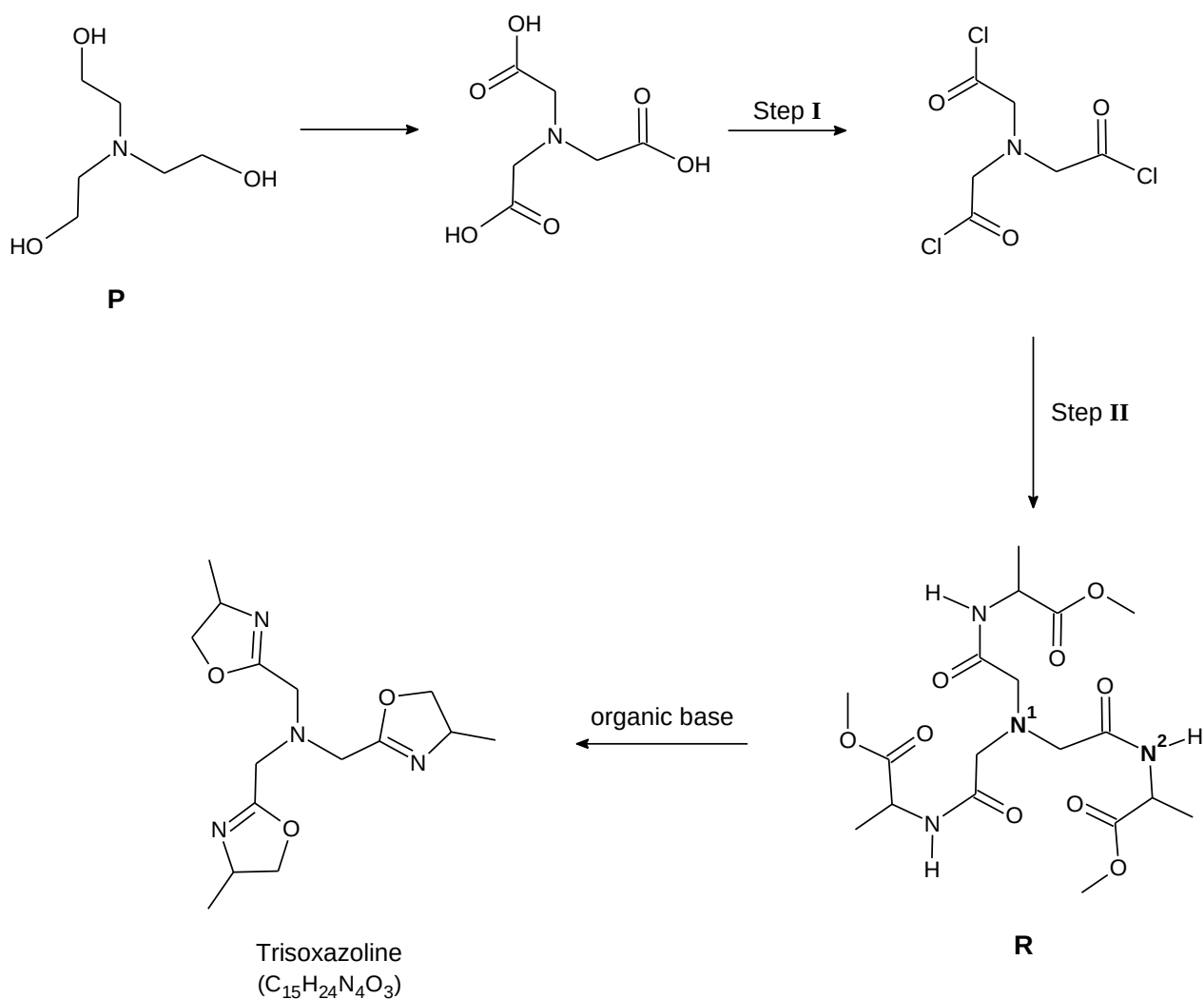
[2]

- (iv) With reference to the *Data Booklet*, predict what will happen when an excess of acidified hydrogen peroxide solution is added to a solution of $\text{Cl}^-(\text{aq})$. [3]

[Total: 17]

molecular structure, they are able to form stable complexes with metals. Metal complexes with trisoxazoline are commonly used as catalyst in organic synthesis.

(a) The reaction scheme below illustrates the synthesis of trisoxazoline.



- (i) Suggest the reagents and conditions required for step **I** and step **II**. [2]
- (ii) Explain the difference in basicity for N^1 and N^2 present in **R**. [2]
- (iii) Propose a 2-step reaction scheme to synthesise compound **P** from ethene. [2]

Trisoxazolines are usually applied to transition metals, such as copper, to form complexes. These complexes are useful in advanced organic synthesis to make specific isomers.

Table 1 shows some properties of the copper complexes, where K_{stab} refers to the stability constant of the complex, e.g.

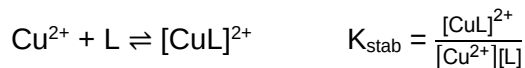
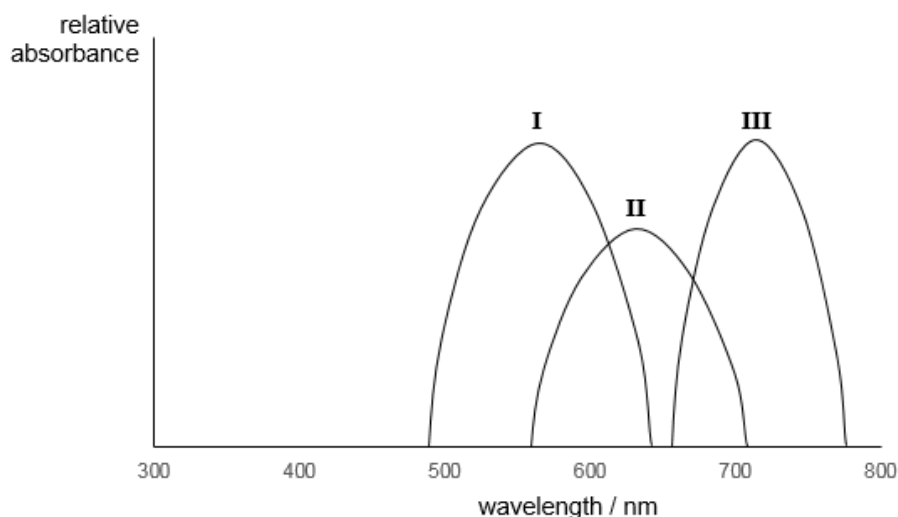


Table 1

Complex	$\log(K_{\text{stab}})$	Colour	Shape of complex
copper(II)-trisoxazoline	12.9	green	distorted tetrahedral
copper(II)-EDTA	18.8	blue	octahedral
copper(II)-en	18.7	violet	octahedral

- (b) Describe the feature of the trisoxazoline molecule that enable it to act as polydentate ligand. Hence, state the type of bond formed between the ligand and the central Cu(II) ion and give the coordination number of the complex. [2]
- (c) (i) Using copper(II)-EDTA as an example, explain why the five d-orbitals are split into two different energy levels. [2]
- (ii) Draw the shapes of the d-orbitals at the higher energy level. [1]
- (iii) Explain why the copper(II) complexes are coloured. [3]
- (iv) The visible absorption spectrum of the three complexes is shown below where each peak represents one complex.

[A visible absorption spectrum is a graph depicting the absorption of radiation by a material over a range of visible light wavelengths.]

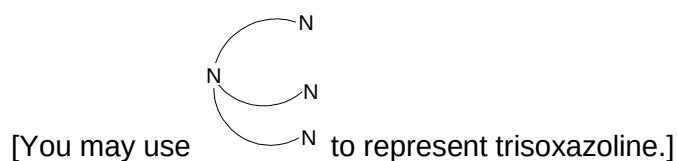


Identify the copper(II) complex responsible for each absorption peak. Hence, arrange the three complexes in order of increasing magnitude of their energy gap.

[2]

- (d) Using Table 1, comment on the spontaneity of forming the respective copper(II) complexes. [2]
- (e) Other than copper(II) ions, it was found that copper(I) ions are also able to form complexes with trisoxazoline.
- (i) Predict and explain the colour of solid copper(I)-trisoxazoline complex. [1]
- (ii) Analysis was done on copper(I)-trisoxazoline complex. The relative molecular mass of the complex was apparently 743.0. Further analysis showed that each copper(I) ion is bonded to three nitrogen atoms.

Propose the structure of copper(I)-trisoxazoline complex.



[1]

[Total: 20]

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Section B is on the next page

Section B

Answer **one** question in this section.

- 4 (a) At r.t.p, 1.2 dm^3 of chlorine gas is reacted with 6.7 g of compound **A**, $\text{C}_9\text{H}_{10}\text{O}$, to give compound **B**. On addition of PCl_5 , **A** gives white fumes. Treatment of **A** with NaBr and concentrated H_2SO_4 under reflux gives a mixture of compounds **C** and **D**. **C** and **D** have the same molecular formula $\text{C}_9\text{H}_{10}\text{Br}_2$ and each contains only *one* chiral centre.

- (i) When NaBr is mixed with concentrated H_2SO_4 , the following observations are made.

- White fumes produced
- Reddish brown liquid formed
- Pungent gas decolourised purple KMnO_4

It is suggested that a displacement reaction occurred first, producing white fumes. Some of the white fumes further react with H_2SO_4 via a redox reaction. Using the above information, write the two corresponding chemical equations for this reaction. [2]

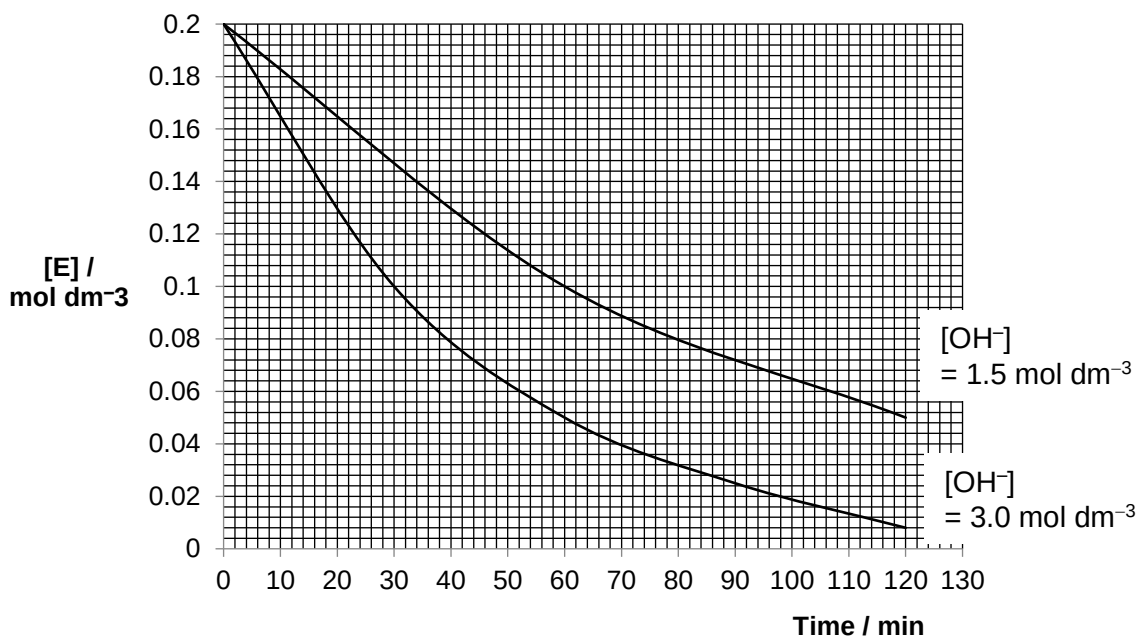
- (ii) Using your answer in (a)(i), deduce the identities of **A** to **D**. [5]

- (iii) Account for the relative quantity of compound **C** and **D** formed. [1]

- (b) **E** is another constitutional isomer of **C** and **D**. A series of chemical experiments were conducted on **E**.

Experiment 1:

The reaction kinetics of **E** with aqueous sodium hydroxide was determined by monitoring the change in concentration of **E** with time.



Experiment 2:

0.2 mol of **D** and **E** were treated separately with boiling aqueous sodium hydroxide. The products from each compound were then acidified with nitric acid and then treated with silver nitrate solution.

The results obtained are shown below:

	Observation upon adding AgNO_3	Mass of ppt / g
D	formation of cream-coloured precipitate	75.12
E	formation of cream-coloured precipitate	37.56

- (i) Using experiment 1, deduce the rate equation. [3]
- (ii) Using Experiment 2 and (b)(i), propose a possible structure of **E** and draw the mechanism for the reaction of **E** with NaOH(aq) . [3]
- (c) Depending on the mechanism, the solvent affects the stability of nucleophile and/or reaction intermediates in the nucleophilic substitution of halogenoalkanes. The table below shows the rate of reaction when different halogenoalkanes and solvents are used.

Solvent	Type	Dielectric constant, ϵ	Relative rates for reaction with OH^-	
			1° RX in $\text{S}_{\text{N}}2$	3° RX in $\text{S}_{\text{N}}1$
CH_3OH	Protic	33	1	4
H_2O	Protic	78	7	150 000
CH_3COOH	Protic	6	1	1
CH_3COCH_3	Aprotic	21	5000	-

Dielectric constant, ϵ , is a measure of the solvent polarity and ability to insulate charge.

- (i) Suggest, using **only** structure and bonding, the difference between *protic* and *aprotic* solvents. [1]
- (ii) Explain the effect of solvent on the relative rate of $\text{S}_{\text{N}}2$ reactions. [2]
- (iii) Explain, with an aid of a diagram, how water increases the rate of $\text{S}_{\text{N}}1$ reaction. [2]
- (iv) Name a solvent, other than those given in the table, that will result in a slower rate for $\text{S}_{\text{N}}1$ than ethanoic acid. [1]

[Total: 20]

- 5 (a) Using only the elements C, H and O, draw the structural formulae of **three** organic compounds, each containing a single carbon atom with an oxidation state of zero, +2 and +4 respectively. [3]
- (b) Hot, concentrated potassium manganate (VII) oxidises several classes of organic compounds to ketones, carboxylic acids or carbon dioxide. By this means, the structures of compounds can be determined. Some compounds are easily oxidised, while others require longer heating.

The following describes some reactions of compounds **F** and **K**, and of their oxidation products.

F, $C_8H_{12}O$ reacts with excess potassium manganate (VII) to produce single organic products, **G**, $C_4H_6O_5$ while **K**, $C_{12}H_{12}$, reacts with the same reagent to produce **H**, $C_{10}H_{10}O_3$. Carbon dioxide is produced in both reactions in a mole ratio of 2 : 1 respectively. During oxidation of **F**, four moles of carbon dioxide were liberated.

Although **F** reacts with potassium manganate (VII), it gives no reaction with potassium dichromate (VI). When 0.10 mol of **F** is reacted with an excess of sodium metal, 1.2 dm^3 of hydrogen is formed, measured at room temperature and pressure.

G reacts with excess concentrated sulfuric acid to give **I**, $C_4H_4O_4$. Upon further oxidation, **I** is found to give **J**, $C_3H_2O_5$, and an inorganic by-product.

H gives a positive iodoform test and dissolves in aqueous sodium hydroxide. Upon further oxidation of **H**, **L**, $C_9H_6O_6$, and a similar inorganic by-product formed from **I** is also produced.

- (i) Deduce the structure of compounds **F** to **L**, explaining the chemistry of the reactions described. [11]
- (ii) State the type(s) of stereoisomerism shown by compound **F** and give one further piece of relevant information about it. [2]
- (iii) Hence, predict the total number of isomers shown by **F**. [1]
- (iv) When compound **K** undergoes prolonged heating with $KMnO_4$ to give **L**, **K** exhibits **two** types of constitutional isomerism.

Name the specific types of isomerism shown by compound **K** and hence, draw the **displayed formulae** of the isomers. [3]

[Total : 20]